Mechanistic Interpretability of Transformer Architectures in Nucleotide Sequence Processing

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***Abstract*—Genomic foundation models already learn useful biological regularities, but most evidence for this claim remains predictive or post hoc rather than mechanistic. At the same time, strict mechanistic interpretability has largely matured in the study of language models rather than nucleotide models. This paper positions small nucleotide transformers as a tractable setting for circuit-level reverse engineering on commodity hardware. We present a tokenization-aware mathematical framework for analyzing nucleotide transformers using QK and OV circuit decomposition, activation patching, and linear probing. We correct the causal restoration metric used for activation patching, formalize a layer-1 theorem for motif detection via QK circuits, and state an approximate extension for deeper layers under a linear decodability assumption. We also report implementation results from a reproducible DNABERT-2 pipeline. Full configured-split residual probes on promoter and splice-site tasks achieve AUROC values of 0.9137, 0.9383, 0.8954, and 0.8847, satisfying the paper's residual-decodability criterion. A full CTCF strict-proof scan over 51,249 GM12878 CTCF sequences produces no head meeting the pre-registered QK-to-motif criterion ($r \geq 0.5$, $p < 0.05$) and no head meeting the matched attention-enrichment criterion ($\rho_h \geq 2.0$). Custom DNABERT forward-hook patching on a promoter-TATA counterfactual identifies one partially restoring head with $\mathrm{PM} = 0.5983$. Thus, the pipeline verifies residual decodability and one promoter-TATA causal-restoration effect, but it does not prove the existence of a strict CTCF motif-detector head. The result clarifies the evidential gap between useful genomic representations and mechanistic motif-detector proofs.**

I. Introduction

The application of transformer architectures to nucleotide sequence analysis has achieved remarkable success in tasks including splice site prediction, transcription factor binding site identification, and regulatory element detection. Models such as DNABERT, Nucleotide Transformer, and SpliceAI demonstrate the predictive value of transformer-based sequence models in genomics [1]–[3]. However, the internal mechanisms by which these models process genomic sequences remain poorly understood.

Mechanistic interpretability provides a framework for reverse-engineering neural networks into formal, algorithmic subgraphs [4], [5]. This approach has yielded significant insights in natural language processing, including the discovery of induction-like behaviors, modular circuits, and causal pathways. The application of these techniques to genomic transformers presents unique opportunities and challenges.

While this topic is not untouched, the specific intersection of strict mechanistic interpretability and nucleotide sequence processing on small, custom-trained models remains underexplored. First, genomic sequence models already exist: DNABERT, GPN, Nucleotide Transformer, and recent interpretable RNA foundation models show that sequence models can learn motifs, gene structure, and related biological regularities from nucleotide data [1], [2], [6], [7]. Second, the most rigorous circuit-level tools in mechanistic interpretability—including QK/OV decomposition, causal tracing, activation patching, and attribution graphs—have been developed primarily in the context of language models [4], [8]–[10]. Third, the field remains bottlenecked by scale: even recent frontier-model analyses report satisfying attribution-graph insight for only about a quarter of the prompts attempted, and long-context attention analysis remains computationally demanding [11], [12]. These constraints make small genomic transformers attractive test beds for exact reverse engineering on consumer hardware.

The fundamental problem addressed in this paper is whether transformers trained on nucleotide sequences develop internal representations that correspond to known biological mechanisms. Specifically, we investigate whether attention heads function as biological motif detectors, whether the residual stream encodes biologically meaningful features, and whether the computational circuits mirror established regulatory mechanisms.

Prior work in genomic deep learning has focused primarily on predictive performance. DeepBind demonstrated that neural networks can learn motif-like position weight patterns for protein binding [13]. SpliceAI showed that deep models with long-range sequence context improve splice site prediction [3]. More recently, DNABERT, GPN, and Nucleotide Transformer extended sequence pretraining to genomic data [1], [2], [6]. However, these studies do not provide circuit-level explanations for how predictions are generated.

The gap between predictive performance and mechanistic understanding has significant implications. Without understanding how models process nucleotide sequences, practitioners cannot trust model predictions in critical applications such as clinical variant interpretation or drug target identification. Furthermore, the inability to interpret model decisions limits the scientific utility of these tools for biological discovery.

This paper makes the following contributions. First, it develops a tokenization-aware mathematical framework for analyzing attention mechanisms in nucleotide transformers,

including formulations for QK and OV circuit decomposition. Second, it corrects the activation patching restoration metric and tightens the theoretical conditions under which a QK circuit can implement motif detection. Third, it defines explicit empirical criteria for claiming causal motif detection, residual-stream decodability, and circuit alignment with biological mechanisms. Fourth, it implements a reproducible DNABERT-2 pipeline that downloads genomic benchmark data, extracts QK/OV circuit matrices, caches frozen residual vectors, trains linear probes, scores JASPAR CTCF motifs, computes QK-to-motif alignment and matched attention enrichment, and performs custom forward-hook activation patching. Fifth, it reports a completed full-data run showing that promoter and splice-site labels are linearly decodable from layer-11 residual states, while the CTCF strict motif-detector criteria fail under the registered thresholds. This negative result is part of the contribution: it prevents the representation-level success from being misreported as a proof of a specific motif-detector head.

II. Related Work

A. Mechanistic Interpretability

Mechanistic interpretability emerged from circuit analysis work on neural networks and has since expanded to transformer architectures, where researchers have identified reusable attention patterns, localized computations, and increasingly detailed computational graphs [4], [5].

Elhage et al. formalized a mathematical framework for understanding transformers, introducing the QK and OV circuit decomposition. This framework separates attention head computation into two components: the query-key circuit determines where attention is directed, while the output-value circuit determines what information is written to the residual stream [4].

Activation patching, introduced in causal tracing work by Meng et al. and systematized by Heimersheim and Nanda, provides a causal intervention method for identifying which components are necessary for specific behaviors [8], [9].

Recent attribution-graph work extends these ideas to frontier models, but also makes their current limits explicit. Anthropic's circuit tracing and biology studies emphasize that successful graphs still explain only a small fraction of total computation and provide satisfying insight for roughly one quarter of prompts attempted [10], [11]. In parallel, Redondo García et al. introduced SPARSE TRACING and the STREAM algorithm, which uses hierarchical pruning to achieve near-linear-time attention analysis and prunes 97–99% of token interactions on long-context reasoning traces [12]. We view these results as important scaling context rather than as evidence that long-context mechanistic interpretability is solved.

B. Genomic Transformers

DNABERT, introduced by Ji et al., applies the BERT architecture to DNA sequences using k-mer tokenization and demonstrates strong downstream performance on promoters, splice sites, and transcription factor binding tasks [1].

GPN shows that unsupervised genomic pretraining can recover gene structure and DNA motifs without direct supervision [6]. Nucleotide Transformer scales genomic pretraining to multiple species and larger model sizes, improving performance on a range of downstream genomics benchmarks [2].

SpliceAI demonstrates the predictive power of long-context sequence modeling for splice site prediction [3]. Beyond DNA, interpretable RNA foundation models such as PlantRNA-FM show that large sequence models can also be paired with motif-level biological analyses in transcriptomic settings [7].

C. Interpretability in Computational Biology

Prior work on interpretability in genomic deep learning has focused primarily on attribution methods and motif visualization. DeepBind produced motif-like filters from predictive models [13]. DNABERT enables nucleotide-level importance visualization and motif inspection [1]. PlantRNA-FM couples a foundation model with biologically meaningful motif analyses in transcriptomes [7].

However, these approaches provide correlational rather than causal explanations. They identify which inputs influence outputs but do not explain how internal computations generate predictions. This gap motivates a study that treats genomic transformers as computational objects whose intermediate states, attention heads, and residual updates can be tested with causal interventions.

III. Methodology

A. Problem Formulation

Let $\mathcal{X} = \{A, C, G, T\}^L$ denote the space of nucleotide sequences of length L. A tokenization map $\tau : \mathcal{X} \to \{1, \ldots, V\}^T$ maps a nucleotide sequence $\mathbf{x}$ to token indices $\mathbf{t} = (t_1, \ldots, t_T)$, where V is the vocabulary size and T is the resulting token length. For single-nucleotide tokenization, $V = 4$ (ignoring special tokens); for fixed k-mer tokenization, $V = 4^k$; and for learned subword schemes such as byte-pair encoding, V is the learned vocabulary size. Let $\mathbf{E}_{\text{tok}} \in \mathbb{R}^{V \times d}$ be the token embedding matrix and $\mathbf{P} \in \mathbb{R}^{T \times d}$ the positional encoding matrix. The initial hidden state is

$$\mathbf{H}^{(0)}_{j,:} = \mathbf{E}_{\text{tok}}[t_j, :] + \mathbf{P}_{j,:}, \quad j = 1, \ldots, T. \tag{1}$$

A bidirectional encoder-only transformer model $f_\theta : \mathcal{X} \to \mathcal{Y}$ maps sequences to predictions, where $\mathcal{Y}$ depends on the task (binary classification for binding sites, multi-class for splice sites, etc.). This bidirectional assumption is important because motif scores such as $s_m(j)$ may depend on tokens spanning a full local window around position j, which is compatible with encoder-style architectures such as DNABERT and Nucleotide Transformer but not with strictly causal masking. The transformer consists of N layers, each containing H attention heads.

Each attention head h in layer l computes:

$$\begin{aligned} \mathbf{Q}^{(l)}_h &= \mathbf{H}^{(l-1)} \mathbf{W}^{Q,(l)}_h, \\ \mathbf{K}^{(l)}_h &= \mathbf{H}^{(l-1)} \mathbf{W}^{K,(l)}_h, \\ \mathbf{V}^{(l)}_h &= \mathbf{H}^{(l-1)} \mathbf{W}^{V,(l)}_h \end{aligned} \tag{2}$$

where $\mathbf{H}^{(l-1)} \in \mathbb{R}^{T\times d}$ is the hidden state entering layer l, and $\mathbf{W}_h^{Q,(l)}, \mathbf{W}_h^{K,(l)}, \mathbf{W}_h^{V,(l)} \in \mathbb{R}^{d\times d_k}$ are learned projection matrices. This tokenization-aware formulation lets us reason explicitly about single-nucleotide, fixed k-mer, and learned subword vocabularies within the same notation.

The attention weights are computed as:

$$\mathbf{A}_h^{(l)} = \mathrm{softmax}\left(\frac{\mathbf{Q}_h^{(l)}(\mathbf{K}_h^{(l)})^T}{\sqrt{d_k}}\right) \tag{3}$$

The head output is:

$$\mathbf{O}_h^{(l)} = \mathbf{A}_h^{(l)}\mathbf{V}_h^{(l)}\mathbf{W}_h^{O,(l)} \tag{4}$$

where $\mathbf{W}_h^{O,(l)} \in \mathbb{R}^{d_k\times d}$ is the output projection matrix.

B. QK and OV Circuit Decomposition

Following Elhage et al., we decompose each attention head into QK and OV circuits. The QK circuit determines the attention pattern:

$$\mathrm{QK}_h^{(l)} = \mathbf{W}_h^{Q,(l)}(\mathbf{W}_h^{K,(l)})^T \tag{5}$$

where $\mathrm{QK}_h^{(l)} \in \mathbb{R}^{d\times d}$.

The OV circuit determines what is written to the residual stream:

$$\mathrm{OV}_h^{(l)} = \mathbf{W}_h^{V,(l)}\mathbf{W}_h^{O,(l)} \tag{6}$$

where $\mathrm{OV}_h^{(l)} \in \mathbb{R}^{d\times d}$.

This decomposition allows us to analyze where attention is directed separately from what information is transmitted.

Definition 1 (Biological Motif Detector)**.** *An attention head h in layer l functions as a biological motif detector for motif m if there exists a nonempty set of relevant query positions $\mathcal{Q}_m$ and a threshold τ such that, for each relevant query $i \in \mathcal{Q}_m$,*

$$\begin{aligned} &\mathbb{E}\left[\sum_{j\in\mathrm{supp}_m(\mathbf{x})} \mathbf{A}_h^{(l)}[i,j] \,\middle|\, \mathrm{supp}_m(\mathbf{x}) \neq \varnothing\right] \geq \tau, \\ &\mathbb{E}\left[\sum_{j\in\mathrm{supp}_m(\mathbf{x})} \mathbf{A}_h^{(l)}[i,j] \,\middle|\, \mathrm{supp}_m(\mathbf{x}) = \varnothing\right] < \tau. \end{aligned} \tag{7}$$

where $\mathrm{supp}_m(\mathbf{x}) \subseteq \{1,\ldots,T\}$ denotes the set of key positions whose associated local sequence windows instantiate motif m. The relevant query position i need not itself lie in $\mathrm{supp}_m(\mathbf{x})$.

C. Activation Patching for Causal Analysis

To establish causal relationships between attention heads and biological predictions, we employ activation patching. Given a clean input $\mathbf{x}_c$ containing a biological motif and a corrupted input $\mathbf{x}_{\text{corr}}$ without the motif, we patch activations from the clean run into the corrupted run.

For attention head h in layer l, the patching operation is:

$$\mathbf{O}_h^{(l),*} = \mathbf{M}\odot\mathbf{O}_h^{(l)}(\mathbf{x}_c) + (1-\mathbf{M})\odot\mathbf{O}_h^{(l)}(\mathbf{x}_{\text{corr}}) \tag{8}$$

where $\mathbf{M} \in \{0,1\}^{T\times d}$ is a mask selecting the token positions and hidden dimensions to patch, and $\odot$ denotes elementwise multiplication.

The patching metric measures the restoration of the clean prediction:

$$\mathrm{PM}(h,l) = \frac{f(\mathbf{x}_{\text{patched}}) - f(\mathbf{x}_{\text{corr}})}{f(\mathbf{x}_c) - f(\mathbf{x}_{\text{corr}})} \tag{9}$$

where f denotes the scalar target quantity of interest (for example, the logit of the positive class) and $f(\mathbf{x}_{\text{patched}})$ is the model output with patched activations. Under this normalization, $\mathrm{PM} = 0$ indicates no recovery, $\mathrm{PM} = 1$ indicates full recovery of the clean effect, and values below 0 or above 1 indicate under-restoration or over-restoration, respectively [9].

D. Linear Probing for Feature Extraction

To analyze what biological features are encoded in the residual stream, we train linear probes on frozen representations. For layer l, the probe is:

$$\hat{y} = \sigma(\mathbf{W}_p\mathbf{h}^{(l)} + b_p) \tag{10}$$

where $\mathbf{h}^{(l)}$ is the hidden state, $\mathbf{W}_p \in \mathbb{R}^{k\times d}$, and σ is the sigmoid function.

In the implemented pipeline, the probe is trained on mean-pooled frozen residual vectors from DNABERT-2 layer 11. The current benchmark tasks are:

- TATA-containing promoter sequences
- non-TATA promoter sequences
- splice donor sequences
- splice acceptor sequences

Probe performance indicates whether the corresponding biological label is linearly decodable from the representation. This is weaker than proving a mechanistic motif detector, but it is a necessary condition for the deeper-layer lemma: if no linear direction encodes the task-relevant biological feature, then the QK alignment condition in the lemma cannot be empirically supported.

E. Mathematical Framework for Biological Circuit Analysis

We formalize the relationship between transformer circuits and biological mechanisms through the following framework.

Definition 2 (Biological Circuit)**.** *A biological circuit is a subgraph of the transformer computation graph consisting of attention heads and MLP neurons that collectively implement a biological function $b : \mathcal{X} \to \{0,1\}$.*

Theorem 1 (Layer-1 Motif Detection via QK Circuits)**.** *Consider a layer-1 attention head operating on $\mathbf{H}^{(0)}$. Let $s_m(j)$ denote a scalar motif score associated with key position j; under single-nucleotide tokenization one has $T = L$, so the token index j coincides exactly with the nucleotide position j. In that setting, one natural choice is*

$$s_m(j) = \sum_{t=0}^{k-1}\sum_{n\in\{A,C,G,T\}} \mathrm{PWM}_m[n,t]\cdot\mathbb{I}[x_{j+t} = n]. \tag{11}$$

If for a fixed query position i there exist constants $a_i \in \mathbb{R}$ and $\gamma > 0$ such that

$$\mathbf{q}_i^T\mathbf{k}_j = a_i + \gamma s_m(j) \tag{12}$$

for all admissible key positions j, then the induced attention distribution is monotonically increasing in $s_m(j)$. In particular, if motif-bearing positions are separated from background positions by a positive margin in s_m, the head is a motif detector for m.

Proof. At layer 1, each key vector depends only on the token embedding and positional encoding at its location. By assumption, the score $\mathbf{q}_i^T\mathbf{k}_j$ is an affine function of the motif score $s_m(j)$. The actual attention logits are

$$\ell_{ij} = \frac{\mathbf{q}_i^T\mathbf{k}_j}{\sqrt{d_k}} = \frac{a_i}{\sqrt{d_k}} + \frac{\gamma}{\sqrt{d_k}} s_m(j). \tag{13}$$

Thus the factor $1/\sqrt{d_k}$ is absorbed into the strictly positive constant $\tilde{\gamma} = \gamma/\sqrt{d_k}$, which preserves monotonicity. Because the exponential and softmax functions are strictly monotone in their inputs, the ordering of the scaled scores is preserved in the attention weights $\mathbf{A}_h^{(1)}[i,j]$. Therefore, positions with larger motif score receive larger attention mass. If every motif-bearing position exceeds every background position by a positive margin, then a threshold on attention mass separates the two classes, which is precisely the motif-detector condition. □

The previous theorem is intentionally restricted to layer 1, where the hidden state is still an explicit function of token and positional embeddings. For deeper layers, an additional decodability assumption is required. This assumption is consistent with the Linear Representation Hypothesis, which posits that semantically meaningful concepts can often be represented as approximately linear directions in activation space [14]. The following lemma makes that dependence explicit and aligns the argument with these linear representation-style assumptions.

Lemma 1 (Approximate Extension Under Linear Decodability)**.** *Suppose that at layer $l-1$ there exists a vector $\mathbf{u}_m$ such that*

$$\mathbf{u}_m^T\mathbf{h}_j^{(l-1)} = s_m(j) + \epsilon_j, \qquad |\epsilon_j| \le \delta, \tag{14}$$

and that the head score satisfies

$$\mathbf{q}_i^T\mathbf{k}_j = a_i + \gamma\mathbf{u}_m^T\mathbf{h}_j^{(l-1)} \tag{15}$$

with $\gamma > 0$. If the motif/background margin in s_m is strictly larger than 2δ, then the ordering between motif-bearing and background positions is preserved and the head remains an approximate motif detector.

Because

$$\begin{aligned}\mathbf{q}_i^T\mathbf{k}_j &= (\mathbf{h}_i^{(l-1)})^T\mathbf{W}_h^{Q,(l)}(\mathbf{W}_h^{K,(l)})^T\mathbf{h}_j^{(l-1)} \\ &= (\mathbf{h}_i^{(l-1)})^T\mathrm{QK}_h^{(l)}\mathbf{h}_j^{(l-1)},\end{aligned} \tag{16}$$

the assumption

$$\mathbf{q}_i^T\mathbf{k}_j = a_i + \gamma\mathbf{u}_m^T\mathbf{h}_j^{(l-1)} \tag{17}$$

implies that the query-specific row of the QK circuit satisfies

$$(\mathbf{h}_i^{(l-1)})^T\mathrm{QK}_h^{(l)} = \mathbf{r}_i^T + \gamma\mathbf{u}_m^T, \tag{18}$$

where $\mathbf{r}_i^T\mathbf{h}_j^{(l-1)} = a_i$ for all admissible key positions j. In the centered-representation or bias-augmented formulation, where the constant offset is represented separately, this reduces to the exact alignment condition

$$(\mathbf{h}_i^{(l-1)})^T\mathrm{QK}_h^{(l)} = \gamma\mathbf{u}_m^T. \tag{19}$$

This equality makes explicit that the QK circuit acts as a linear transformation isolating the motif direction seen by the query.

Proof. For any motif-bearing position j_m and background position j_b,

$$\mathbf{q}_i^T\mathbf{k}_{j_m} - \mathbf{q}_i^T\mathbf{k}_{j_b} = \gamma\big(s_m(j_m) - s_m(j_b) + \epsilon_{j_m} - \epsilon_{j_b}\big). \tag{20}$$

The perturbation term lies in $[-2\gamma\delta, 2\gamma\delta]$. Hence a clean separation margin greater than 2δ preserves the sign of the score difference, so softmax preserves the motif/background ordering under the same separation criterion. □

Proposition 1 (Residual Stream Encoding)**.** *The residual stream at layer l encodes biological feature f if there exists a linear projection $\mathbf{w}_f$ such that:*

$$\mathbf{w}_f^T\mathbf{h}_i^{(l)} = f(x_{i-w:i+w}) + \epsilon \tag{21}$$

where w is the context window and ϵ is approximation error.

F. Experimental Setup

We propose instantiating the framework on three genomic transformer families:

DNABERT: a 6-layer transformer with 12 attention heads per layer and fixed 6-mer tokenization [1].

Nucleotide Transformer: a larger multi-species transformer with learned genomic tokenization [2].

Custom Splice Transformer: a small model trained specifically for splice site prediction with single-nucleotide tokenization, intended as the primary tractable setting for exhaustive circuit analysis.

For each model, the analysis requires:

- Attention patterns from all heads and layers
- Hidden states at each layer
- QK and OV circuit matrices
- Optional MLP activations for follow-up circuit tracing

The current repository implementation instantiates the framework on `zhihan1996/DNABERT-2-117M`. The run uses CUDA when available, exports QK and OV matrices for layers 0, 5, and 11, and caches mean-pooled residual vectors from the same layers. Because the installed TransformerLens backend does not fully support DNABERT-2's custom BERT-style implementation, the current run uses a Hugging Face forward-hook adapter for residual extraction, direct weight extraction for QK/OV matrices, low-rank W_Q/W_K factor scoring for QK alignment, and custom PyTorch forward hooks for DNABERT attention-output patching. This preserves the residual-probing, circuit-matrix, QK-alignment, enrichment, and custom causal-intervention parts of the framework even when TransformerLens-style per-head patching utilities are unavailable.

Evaluation can be performed on benchmark datasets including:

- ENCODE ChIP-seq for transcription factor binding
- GENCODE annotations for splice sites
- JASPAR motifs and consensus sequences
- Synthetic sequences with implanted or deleted motifs

IV. Evaluation Protocol and Success Criteria

Because probe results alone are not sufficient for mechanistic proof, this section specifies the evidence thresholds required to support the paper's strongest claims.

A. Candidate Head Screening

A head is promoted from exploratory inspection to mechanistic analysis only if it satisfies the criteria in Table I. These thresholds should be fixed before inspecting held-out results to reduce post hoc threshold selection.

TABLE I
Pre-Registered Evidence Criteria for Candidate Motif Circuits

Claim	Statistic	Threshold
Motif localization	Attention enrichment ratio $\rho_h = \mathbb{E}[a_{\text{motif}}]/\mathbb{E}[a_{\text{bg}}]$	$\rho_h \geq 2.0$
Motif ranking	Correlation of QK score with motif score	$r \geq 0.5$, $p < 0.05$
Causal importance	Patching metric $\text{PM}(h, l)$	$\text{PM} \geq 0.5$
Strong causal importance	Same as above	$\text{PM} \geq 0.75$
Residual decodability	Probe AUROC or macro-F_1 above baseline	≥ 0.80 or $+0.10$
Robustness	Matched negative controls	No matched effect

Here a_{motif} denotes attention mass assigned to motif-support positions and a_{bg} denotes matched background attention mass. Multiple testing should be controlled with permutation or false-discovery-rate corrections.

B. Causal Intervention Protocol

Clean and corrupted sequence pairs should differ only in the presence, absence, or mutation of a target motif. Each patching experiment should be accompanied by token-shuffled controls, norm-matched ablations, and replication across random seeds. A single high patching score is insufficient unless it co-occurs with motif-localized attention and a biologically interpretable OV readout.

C. Representation Probing Criteria

Claims about residual-stream encoding require probes to outperform simple baselines such as position-only or GC-content-only classifiers. Probe results should be reported with calibration, class balance, and random-label controls so that linear separability is not confused with dataset leakage.

D. Circuit Structure Criteria

A QK circuit is treated as biologically meaningful only when three signals converge: localized attention on motif-support positions, causal restoration under activation patching, and an OV readout that measurably shifts downstream logits or probe-relevant directions. This multi-criterion requirement reduces the risk of overinterpreting attention patterns alone.

V. Project Implementation and Results

We implemented the proposed pipeline in the MINTS repository as a one-command reproducibility workflow. The pipeline ingests the revised Nucleotide Transformer downstream task dataset, downloads ENCODE GM12878 CTCF artifacts, prepares GRCh38-derived CTCF peak sequences, loads DNABERT-2, exports selected QK/OV circuit matrices and W_Q/W_K factors, caches frozen residual-stream vectors, trains logistic regression probes, computes CTCF QK-to-motif alignment, computes matched attention enrichment, and runs custom forward-hook activation patching. The root run summary is written to `results/pipeline_run.json`; the QK-alignment, enrichment, patching, and heatmap artifacts are written under `results/qk_alignment/`, `results/enrichment/`, `results/patching/`, and `results/figures/`.

The latest completed run was executed on April 13, 2026 and used `zhihan1996/DNABERT-2-117M` on CUDA through the Hugging Face forward-hook backend. It completed from the repository entrypoint `python main.py` in 3,261.229 seconds, with 2,739.109 seconds spent in the strict proof step. It exported QK and OV matrices for layers 0, 5, and 11. Each exported layer has 12 heads with $d_{\text{model}} = 768$ and $d_{\text{head}} = 64$. For residual probing, the completed run used all rows in the configured train and test splits: 5,062/212 train/test examples for the TATA promoter task, 30,000/1,372 for the non-TATA promoter task, and 30,000/3,000 for each splice-site task. The positive class rates were close to balanced in all train and test splits, ranging from 0.4917 to 0.5028.

TABLE II
Full Configured-Split Layer-11 Residual Probe Results from the Implemented Pipeline

Task	AUROC	AUPRC	Acc.	Test n
promoter_tata	0.9137	0.9241	0.8349	212
promoter_no_tata	0.9383	0.9475	0.8550	1372
splice_sites_donors	0.8954	0.9049	0.8230	3000
splice_sites_acceptors	0.8847	0.8954	0.8090	3000

These results show that the selected biological task labels are linearly decodable from the frozen residual stream at layer 11 under the full configured-split run. Under the success criteria in Table I, all four tasks meet the residual-decodability threshold of AUROC ≥ 0.80. This is a genuine positive result for the representation-level part of the theory: it supports the assumption that biologically meaningful task information is present in approximately linear directions in the residual stream.

The strict CTCF scan used all 51,249 prepared GM12878 CTCF peak sequences. JASPAR CTCF motif scoring produced 256,918 motif-support tokens and 1,843,874 finite token positions for QK-to-motif correlation. Table III summarizes the results against the criteria from Table I.

The CTCF result is a negative strict-proof result, as visualized in Fig. 1(a–b). Although several correlations are statistically nonzero because the scan contains more than 1.8 million finite token positions, the largest effect size, $r = 0.2119$, is well below the pre-registered $r \geq 0.5$ threshold. Similarly, the largest matched enrichment ratio, $\rho_h = 1.0209$, is close to background and far below the $\rho_h \geq 2.0$ threshold. Thus, none of the tested CTCF heads is promoted to a strict motif-detector candidate.

The promoter-TATA patching result in Fig. 1(c) is a separate positive causal signal, not a complete motif-detector proof. The executed clean/corrupted pair was `chr20:257674-257974|1`, where the clean subsequence `TATAAA` at character span $[20, 26)$ was replaced by `GCGCGC`. The scalar target was the trained promoter-TATA residual-probe decision function. The best intervention, layer 7 head 8, restored 59.83% of the clean-minus-corrupted effect. This crosses the causal-importance threshold $\mathrm{PM} \geq 0.5$ but not the strong-causal threshold $\mathrm{PM} \geq 0.75$. Because the CTCF QK and enrichment tests produced no passing candidate heads, and because this patching run targets a promoter-TATA pair rather than a CTCF pair, the completed run does not prove a strict CTCF biological motif-detector head.

The implementation also clarifies a practical limitation of applying language-model interpretability tools directly to genomic BERT-style models. DNABERT-2 can be loaded and analyzed through residual hooks and direct weight extraction, but the current TransformerLens environment does not expose all per-head attention-output tensors needed for direct `get_act_patch_attn_head_out_by_pos` experiments. For that reason, the repository uses custom DNABERT forward hooks for the executed head-output patching experiment while retaining TransformerLens wrapping when the backend permits it.

VI. Interpretive Hypotheses

The implementation results in Section V motivate but do not prove the interpretive hypotheses below. The residual-probe AUROCs indicate that promoter and splice-site labels are linearly decodable from the frozen representation, which is consistent with the deeper-layer decodability lemma. The completed CTCF QK/enrichment scan also shows that this decodable biological information should not be assumed to reside in simple motif-local QK heads: under the registered criteria, no tested CTCF head passed the QK-alignment or matched-enrichment threshold. The remaining question is whether the useful information is carried by different layers, MLP features, distributed residual directions, task-specific non-CTCF circuits, or motifs not captured by the single JASPAR CTCF PWM test.

A. Expected Mechanism of Motif Detection

Early layers, especially under single-nucleotide or fixed k-mer tokenization, should implement local pattern matching because token identity remains directly accessible in the residual stream. Under this hypothesis, motif-sensitive heads will have QK scores that track sequence match scores and OV circuits that write task-relevant evidence into downstream features.

For motif m of length k, one expects the candidate detector score at position i to satisfy an approximation of the form

$$s_i \approx \sum_{t=0}^{k-1} \mathbf{w}_m^T \mathbf{e}_{x_{i+t}} + b_m, \tag{22}$$

where $\mathbf{w}_m$ is a motif-aligned direction and $\mathbf{e}_{x_{i+t}}$ is the embedding associated with nucleotide x_{i+t} or the token covering that position.

B. Expected Circuit Organization

For splice-site tasks, a plausible circuit organization consists of multiple stages:

Early layers: local detectors identify donor, acceptor, promoter, or composition signals.

Intermediate layers: context integrators combine local motifs with surrounding sequence statistics.

Late layers: aggregators pool evidence across the sequence for final prediction.

This is a hypothesis about circuit organization, not an empirical claim about DNABERT-2. In the completed CTCF scan, layers 0, 5, and 11 did not contain a head satisfying the strict CTCF motif-detector thresholds.

The full configured-split residual-probe results support only the weakest part of this organization: late residual states contain enough information for linear classification of donor and acceptor labels. The custom promoter-TATA patching result adds evidence that one head can causally restore part of a task-relevant probe signal for a TATAAA mutation. However, the run still does not identify a complete local donor, acceptor, promoter, or CTCF detector circuit satisfying all QK, enrichment, and causal criteria.

C. Expected Alignment with Biological Knowledge

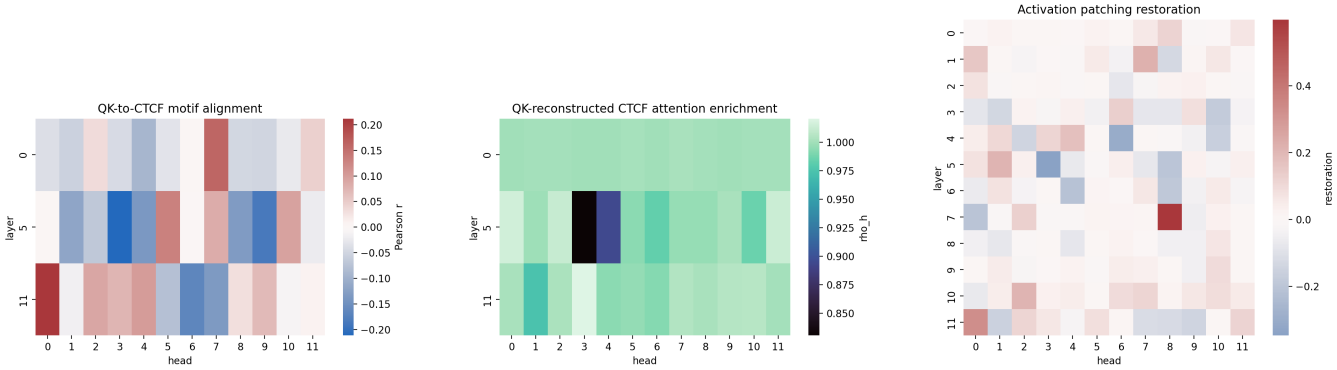
Promoter or splice models may recover canonical motifs such as TATA-like promoter elements or GU–AG splice signals, but alignment with known biology should be treated as a testable outcome rather than presumed ground truth. Deviations are also informative: they may reflect data bias, alternate regulatory strategies, or tokenization artifacts.

D. Failure Cases

Polysemanticity and feature superposition may cause a single head to respond to multiple motifs or to context-dependent composites rather than a single biological primitive. Attention can also be gated by surrounding context, and some biological variables may remain sufficiently nonlinear or distributed that linear probes fail even when the model uses them.

TABLE III
STRICT MECHANISTIC EVIDENCE SUMMARY FROM THE COMPLETED DNABERT-2 RUN

Evidence channel	Executed test	Best observed result	Criterion met?
Residual decodability	Layer-11 logistic probes on four benchmark tasks	AUROC range 0.8847–0.9383	Yes
CTCF QK alignment	36 heads across layers 0, 5, 11; Pearson correlation with JASPAR CTCF token scores	Best: layer 11 head 0, $r = 0.2119$, $p \approx 0$, $n = 1{,}843{,}874$	No
CTCF attention enrichment	36 heads across layers 0, 5, 11; matched motif/background attention mass	Best: layer 11 head 3, $\rho_h = 1.0209$, 256,918 support tokens	No
Promoter-TATA patching	144 layer/head interventions on a TATAAA $\rightarrow$ GCGCGC counterfactual	Best: layer 7 head 8, $\mathrm{PM} = 0.5983$	Partial
Strong patching	Same promoter-TATA interventions	No head reached $\mathrm{PM} \geq 0.75$	No
Strict CTCF motif detector	Same head must pass CTCF QK alignment, CTCF enrichment, and causal restoration	No CTCF QK or enrichment candidate heads	No



(a) CTCF QK-to-motif Pearson correlations. (b) CTCF matched attention-enrichment ratios. (c) Promoter-TATA activation-patching restoration.

Fig. 1. Mechanistic evidence heatmaps from the completed MINTS run. Panels (a) and (b) show the negative CTCF strict-proof result: no tested head reaches the registered QK-alignment threshold $r \geq 0.5$ or matched-enrichment threshold $\rho_h \geq 2.0$. Panel (c) shows the separate promoter-TATA patching experiment, where layer 7 head 8 is the strongest partial-restoration head with $\mathrm{PM} = 0.5983$.

VII. LIMITATIONS

This study has several limitations. First, it is intentionally scoped to small and medium-sized models because current mechanistic tools do not yet provide exhaustive explanations of frontier systems. Recent attribution-graph analyses report satisfying insight for only about a quarter of prompts attempted and emphasize that even successful case studies recover only a subset of total computation [10], [11].

Second, activation patching establishes causal importance but does not fully characterize the computational role of each head. Additional techniques such as path patching or full attribution-graph recovery are still needed for complete circuit characterization [8]–[10].

Simple co-activation statistics are not sufficient for this purpose. They can reveal which features tend to fire together, but they cannot distinguish correlation from causation or determine the direction of information flow between features. Single-feature causal patching is therefore required to test candidate circuit edges directly, yet scaling such interventions from isolated features to full attribution graphs across thousands of base pairs remains computationally prohibitive.

Third, long genomic contexts remain challenging. Recent sparse tracing work suggests a path forward by achieving near-linear attention analysis and pruning 97–99% of token interactions on long traces [12], but this capability has not yet been validated for genomic sequences or for the specific motif analyses proposed here.

Fourth, biological annotations are incomplete. Many regulatory elements remain uncharacterized, and any motif-centered analysis is therefore limited to partially observed biology.

Fifth, the linear probing assumption that features are linearly encoded may not hold for all biological concepts. Nonlinear probing methods may reveal additional encoded features.

Sixth, the empirical results reported here show residual decodability on four benchmark tasks, successful QK/OV export for selected DNABERT-2 layers, a completed CTCF QK-to-motif scan, a completed matched attention-enrichment scan, and a completed custom promoter-TATA patching heatmap. The strongest mechanistic claim in the paper—the existence of a specific strict CTCF biological motif-detector head—is not supported by these results. The negative CTCF result is informative, but it is not a universal impossibility theorem: it applies to the selected DNABERT-2 model, the exported layers 0, 5, and 11, the prepared GM12878 CTCF intervals,

the JASPAR MA0139.1 scoring procedure, and the registered thresholds.

VIII. Conclusion

This paper reframes the mechanistic study of nucleotide transformers as a precise, underexplored problem at the intersection of genomic foundation models and circuit-level interpretability. It provides a mathematically consistent framework for testing whether attention heads act as motif detectors and whether residual states encode biologically meaningful features, and it reports a completed full-data implementation of that framework on DNABERT-2 for the configured tasks.

The corrected patching metric, tokenization-aware formulation, layer-1 theorem, and deeper-layer lemma together specify what would count as credible evidence. The implemented pipeline achieves the residual-decodability part of this evidence stack: biological labels for promoter and splice-site tasks are linearly decodable from layer-11 residual representations, with full configured-split AUROC values ranging from 0.8847 to 0.9383. It also completes the CTCF QK/enrichment tests and the promoter-TATA custom patching test. The strict CTCF motif-detector proof fails under the registered thresholds: no tested head reaches the QK alignment or attention-enrichment criterion. The promoter-TATA patching experiment does identify one partially restoring head, layer 7 head 8 with $\mathrm{PM} = 0.5983$, but this is not sufficient to establish a complete motif-detector circuit.

Future work should extend QK/enrichment scans beyond layers 0, 5, and 11; run motif-specific patching for CTCF as well as promoter and splice motifs; add OV readout analysis and matched negative controls; compare tokenization schemes directly; and determine whether sparse long-context tracing can recover circuits over promoter, enhancer, and splicing windows at genomic scale.

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